The Meridional Heat Flux across the SAMBA Section

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ABSTRACT

Oceanic circulation is established, in the Atlantic Ocean, through the AMOC. Considering a transect throughout the basin, it can be analyzed as a closed box crossed by meridional heat transport through different water masses at different depths. This paper analyzes the Meridional Heat Transport (MHT) in the Atlantic Basin across the SAMOC Basin-Wide Array (SAMBA) transect, from 1958 to 2023, using data provided by the "OFES" ocean model. The zonally integrated transport was calculated to separate the basin into three sectors: Western Margin, Eastern Margin, and the Interior of the Basin. The depth-based integrated transport followed the same principle, dividing the basin into the Upper Layer, Intermediate Layer, and Deep Layer. The MHT shows a mean of 0.30 PW northward, with an increasing trend of 0.045 PW/decade. The seasonal mean shows an increase in northward transport during the winter and a decrease during the summer, peaking in June at 0.53 PW and reaching a minimum in February at 0.16 PW, which corroborates previous findings. The Western Margin exhibits a southward heat transport with a decreasing trend towards the present, showing a strong correlation with the total transport of the basin. Conversely, the Eastern Margin and the Interior of the Basin show a northward transport with no sign of correlation. The MOC intensity, calculated by integrating up to the maximum northward transport, is strongly correlated with the total MHT across the basin. The Upper Layer is also strongly correlated, showing an increasing trend towards the present. The intermediate layer shows a moderate negative correlation with the MHT, while the deep layer shows a moderate-to-good negative correlation. The model data indicate that the upper layer is primarily responsible for the heat transport through the section, and that the maximum northward transport is the main driver of these changes. Considering this, the strength of the boundary currents is what controls the heat changes throughout the basin.

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1 Introduction

1.1 Contextualization

Ocean dynamics are a result of the difference between solar radiation incidence, which is responsible for wind distribution, and the upwelling-downwelling processes of water masses due to differences in temperature and salinity across different waters (Garzoli, Baringer, et al. 2013). These processes, linked to the Earth's rotation, are responsible for the global heat transport known as "thermohaline circulation". In the Atlantic Ocean, the thermohaline circulation is established by the Atlantic Meridional Overturning Circulation (AMOC) (Rahmstorf 2006).

AMOC's temporal and spatial analysis is crucial due to its location: the flow is derived from the Pacific Ocean through the Drake Passage, flowing into the Indian Ocean around the Cape of Good Hope, and interacting with the entire Atlantic Basin (Chidichimo et al. 2023). To simplify the analysis, the basin can be described as a closed-box mechanism (more specifically, as a compartment with only one entrance and one exit) where the AMOC functions as the sole oceanographic mechanism responsible for transporting heat from the equator to polar regions and vice versa (Pond and Pickard 1983; Chidichimo et al. 2023).

Meridional Heat Transport (MHT) is based on the conservation of mass, in which the upper layer tends to transport heat northwards, compensating for the southwards transport carried by the intermediate layer. This process is mainly driven by the North Atlantic Deep Water (NADW) as it exits the Atlantic and interacts with water masses from other basins. In this circulation, deep waters also flow northwards due to Antarctic Bottom Water (AABW) cells (Pond and Pickard 1983; Baringer and Garzoli 2007). Quantifying these fluxes is essential to understanding the Earth's energy-climate balance, as well as for the modeling and forecasting of global climate mechanisms (Garzoli, Baringer, et al. 2013).

Functioning as a climate regulator, the ocean also stores anthropogenic heat and carbon, making the AMOC directly vulnerable to climate change scenarios. Under these scenarios, with global temperature projected to increase by 2°C, the AMOC may have already shown signatures of this warming over recent decades. This influences not only global circulation but also biogeochemical cycles essential to human life, such as nutrient distribution and the availability of natural resources (IPCC 2022; Talley, Reid, and Robbins 2003).

Regarding circulation exclusively, transport varies in depth and longitude based on differences between water mass characteristics, such as salinity and temperature. In the Atlantic, the basin margins differ due to colder, less salty Pacific waters entering the western margin, while the eastern margin receives waters originating from the Agulhas Retroflection (Chidichimo et al. 2023; Baringer and Garzoli 2007).

The South Atlantic basin is also highly relevant due to the currents that carry waters from these interoceanic exchanges, such as the Benguela Current, which enters the basin from the Agulhas Retroflection. Along the western margin, the Brazil Current is essential to the Meridional Heat Flux across the South Atlantic Basin and poleward, where it meets the Malvinas Current at the Brazil-Malvinas Confluence, exchanging sensible heat with water masses near the Antarctic Continent

(Biastoch et al. 2009; Peterson and Stramma 1991; Gordon and Greengrove 1986; Garzoli and Matano 2011). These processes not only vary spatially but also exhibit temporal cycles, responding to planetary variations and changing according to seasons and wind patterns (Meinen et al. 2013).

1.2 Paper Goals

The main goal of this paper is to conduct a temporal and spatial analysis of the AMOC by evaluating temperature advection based on numerical models, estimating its variability both as a whole and within specific compartments, and investigating its cycles. The secondary goals are to compare these findings with external datasets and published literature to identify correlations with the observed signals.

2 Material and Methods

2.1 Study Area

The analyzed section is related to the SAMOC (South Atlantic Meridional Overturning Circulation) project, an international cooperation established by a global observation network shared among different universities. The chosen transect is the SAMOC Basin-wide Array (SAMBA) section, located at 34.5°S latitude, where the MHT entering or exiting the basin will be calculated (Figure 1).

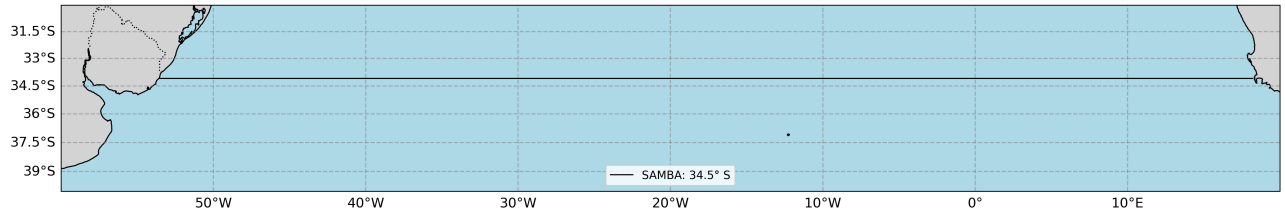


Figure 1: Schematic representation of the Atlantic Basin, with the SAMBA transect highlighted in black.

2.2 Data Availability

Temperature and velocity data across different depths were obtained from the Ocean General Circulation Model for the Earth Simulator (OFES) platform. Developed by the Japan Agency for Marine-Earth Science and Technology (JAMSTEC), OFES is a high-resolution numerical ocean model that solves equations based on a hydrostatic approximation of the ocean, utilizing a centered finite-difference scheme to calculate advection (Masumoto 2010).

2.3 Methodology

The MHT across the transect is calculated as the heat flux perpendicular to the zonal section, following the methodology described by Baringer and Garzoli (2007). The calculation is formulated as follows:

$$\mathrm{MHT} = \iint_A \rho \, C_p \, T \, v \, \mathrm{d}x \, \mathrm{d}z \tag{1}$$

where ρ is the density of the medium, C_p is the specific heat capacity of water, and T and v are, respectively, the temperature and the meridional velocity component of the fluid at the point of interest, while x and z represent the zonal and vertical spatial coordinates. The vertical grid spacing is variable and provided directly by the model output for each level, whereas the horizontal grid spacing is based on the model resolution (1/10°).

To analyze the MHT variation, integration was performed over x and z, covering all months from January 1958 to December 2023. The seasonal breakdown for the Southern Hemisphere was defined as follows: December, January, and February (DJF) for summer; March, April, and May (MAM) for autumn; June, July, and August (JJA) for winter; and September, October, and November (SON) for spring. For the 1958 summer, since data for December 1957 were unavailable, the mean meridional heat transport was calculated using only the values obtained in January and February 1958.

The heat transport carried by each component of the Atlantic circulation was calculated to evaluate the relevance of different water masses and currents to oceanic heat transport. Accordingly, the data were separated based on northward and southward transport, as well as the geographical location of each transport mechanism. For the vertical layers, the depth separation was established through the following calculation, followed by a cumulative sum:

$$\mathrm{AMOC}(z_j) = \frac{1}{N} \sum_{n=1}^{N} \left[\sum_{k=0}^{j} \left(\sum_{i=1}^{M} v_n(x_i, z_k) \, \Delta x \, \Delta z_k \right) \right] \tag{2}$$

The separation into different regions was based on the integration of meridional velocity across each longitude to estimate the zonal contribution to the transport. The calculation is formulated as follows:

$$\overline{T}(x_i) = \frac{1}{N} \sum_{n=1}^{N} \sum_{j=1}^{P} v_n(x_i, z_j) \, \Delta x \, \Delta z_j \tag{3}$$

Based on these defined regional boundaries, the MHT was calculated in its standardized form, as presented in equation (1).

3 Results and Discussion

3.1 Meridional Heat Transport through the basin

The total MHT calculation evidences a northward transport, with a mean of 0.30 PW and an increase of 0.045 PW per decade. Decadal means were 0.11 PW for the 1960s, 0.23 PW for the 1970s, 0.32 PW for the 1980s, 0.38 PW for the 1990s, and 0.39 PW and 0.38 PW for the 2000s and 2010s, respectively. A certain stabilization is observable near the 1990s, where values remain around the same average. The annual means present a similar behavior, showing an upward trend towards the present (Figure 2).

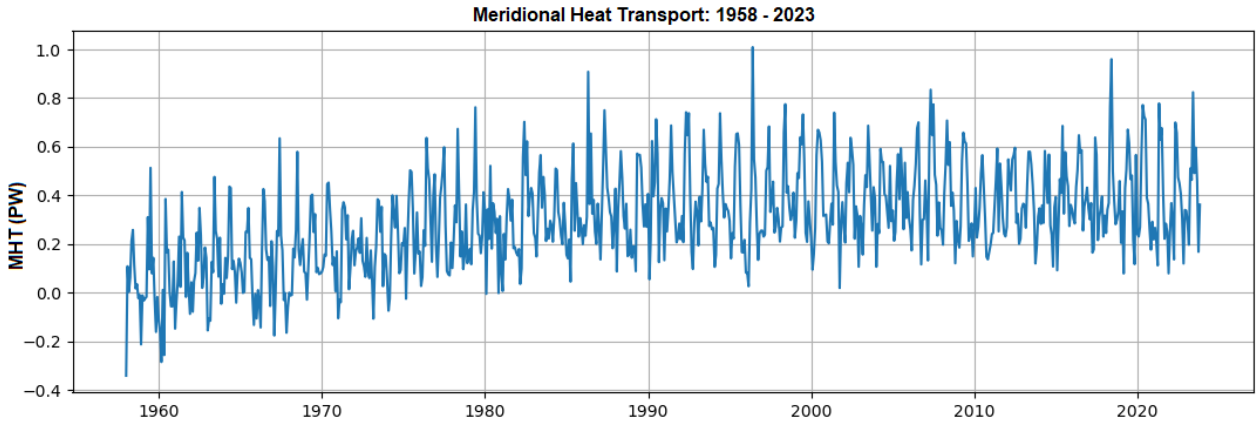


Figure 2: Total MHT (in blue) calculated based on the monthly means from the OFES model.

Considering equation (1), this increase and variability towards the present may be controlled by variations in temperature, velocity, or both within the basin. Most published papers, when used for comparison, lack the temporal coverage observed in the OFES model data because they work directly with *in situ* data and use model outputs merely for correlation within limited periods. In other sections, this issue persists, leading to partial conclusions regarding interannual variability and reinforcing the need for continuous monitoring programs. Thus, for the data analysis in this study, the most effective approach lies in analyzing seasonal variations and compartments associated with transport mechanisms (currents and water masses).

3.2 Seasonal Variability

Regarding interannual variations, all seasons also exhibit an increasing trend in northward transport towards the present. The winter season presents the highest northward transport values, whereas summer shows the lowest. Spring and autumn exhibit intermediate and similar behaviors (Figure 3). The monthly means also reflect these patterns, with the highest average occurring in June (0.53 PW) and the minimum in February (0.16 PW), highlighting the higher means observed during winter (Figure 4).

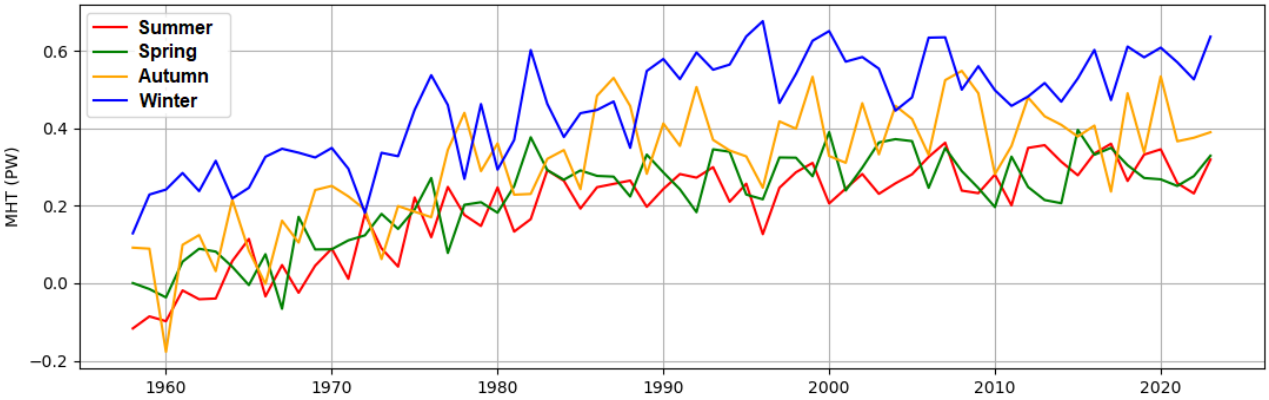


Figure 3: Monthly means by season: summer (red), autumn (orange), spring (green), and winter (blue).

According to Baker et al. (2023), seasonal variation is caused by differences in transport along

continental margins, with variations on the western margin being dominant and showing maximum northward transport during August. Despite the difference in the specific peak months, both occur during the Southern Hemisphere winter (Northern Hemisphere summer). According to Dong et al. (2009), the variability of this seasonal change is mostly controlled by geostrophic and Ekman layer transport (61% and 51%, respectively). It is therefore associated with the velocity component rather than temperature, confirming, in these cases, the winter maximum and summer minimum.

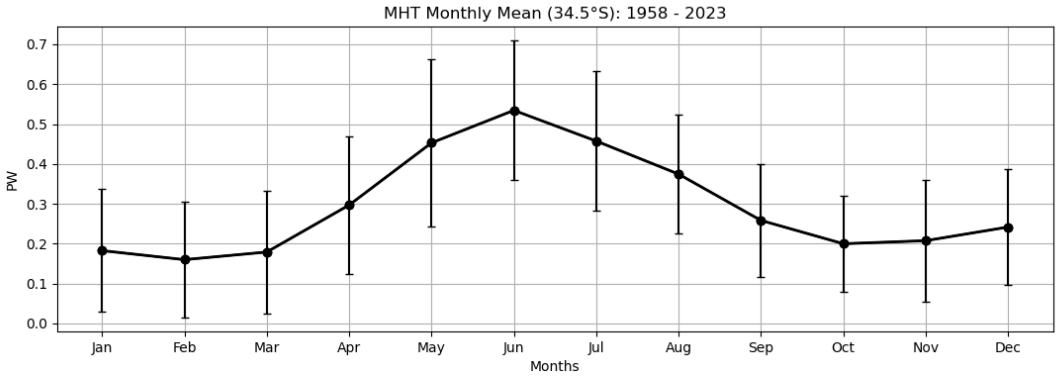


Figure 4: Means for each month relative to the data available in the OFES model from January 1958 to December 2023.

3.3 Zonal Variability

Regional compartmentalization was established based on the mean flow velocity and its direction. The western boundary (comprising the Brazil Current) transports volume southward. The eastern boundary (comprising the Benguela Current and the Agulhas Retroflection) exhibits predominantly northward transport, influenced by its proximity to the bathymetric contour of the Walvis Ridge. The remaining region constitutes the geostrophic interior. Consequently, the western margin is defined as the heat flux west of 49°W, the basin interior as the flux between 49°W and 6°E, and the eastern margin as the region east of 6°E (Figure 5).

The western margin region features predominantly southward transport, with a slight decrease in southward transport over the decades—a behavior similar to that observed in the total MHT calculation. Decadal values were −1.04 PW for the 1960s, −0.93 PW for the 1970s, −0.80 PW for the 1980s, −0.82 PW for the 1990s, −0.77 PW for the 2000s, and −0.80 PW for the 2010s. Conversely, the eastern margin shows a slight decrease in northward transport, with values of 0.72 PW for the 1960s, 0.65 PW for the 1970s, 0.71 PW for the 1980s, 0.64 PW for the 1990s, 0.65 PW for the 2000s, and 0.62 PW for the 2010s. The interior displays an oscillatory behavior without a clearly defined pattern, recording 0.43 PW for the 1960s, 0.50 PW for the 1970s, 0.42 PW for the 1980s, 0.57 PW for the 1990s, 0.52 PW for the 2000s, and 0.56 PW for the 2010s (Figure 6).

Based on the trends observed on the western boundary, a linear correlation analysis was performed between the time series of the different basin regions and the total heat transport. The eastern margin and the basin interior showed virtually no correlation, with values close to 0, whereas the western margin exhibited a moderate positive correlation of 0.468 (Figure 7). This indicates that

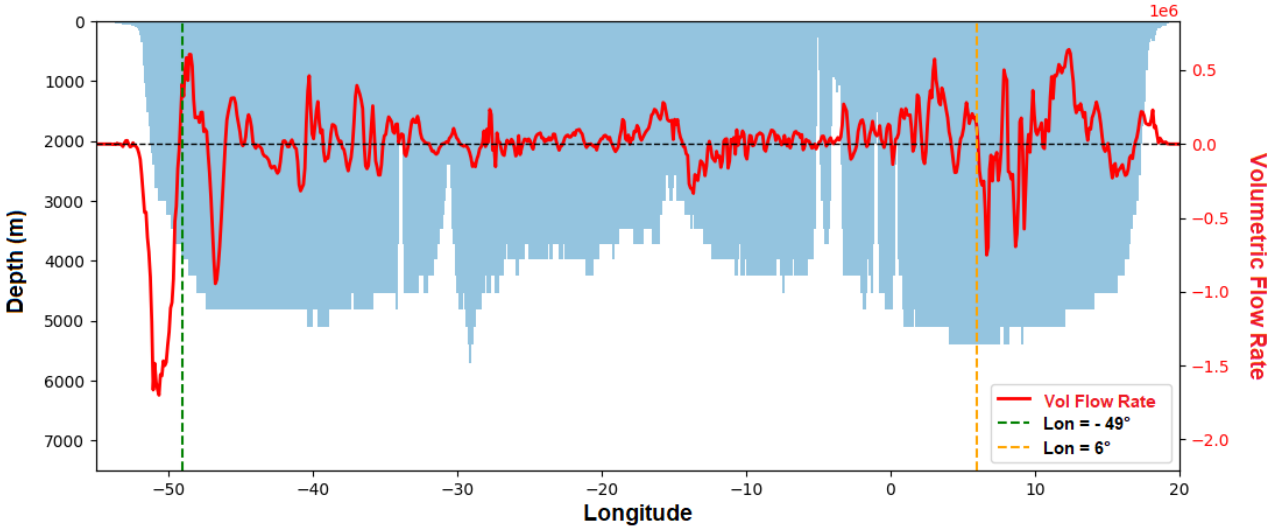


Figure 5: Flow velocity (in red) across the 34.5°S section, varying by longitude. The red line represents northward or southward transport for the longitude overlapping the observed section.

the western margin controls a significant portion of the circulation, corroborating the signal observed by Dong et al. (2009).

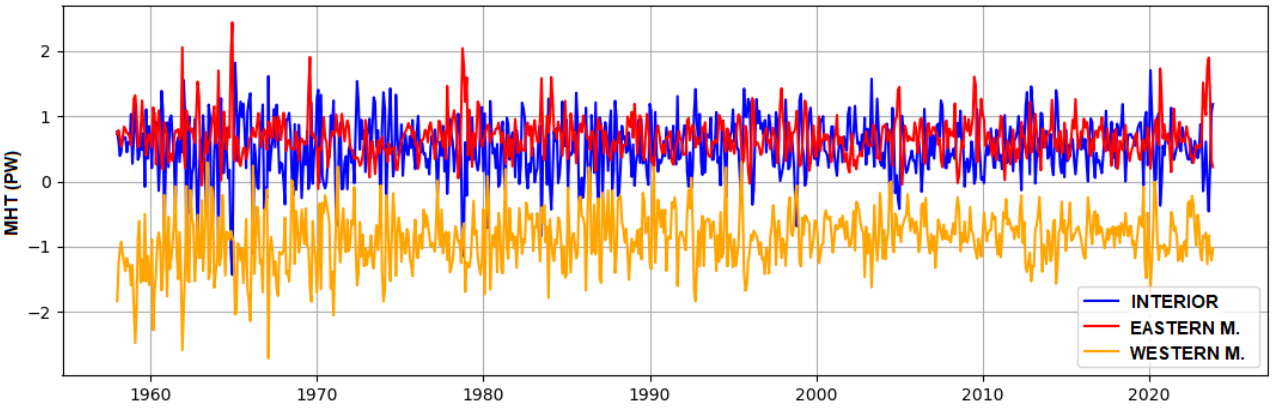


Figure 6: MHT calculation based on the separation of the different regions of the Atlantic basin: Western Margin (yellow), Interior (blue), and Eastern Margin (red).

3.4 Depth Variability

The MOC intensity—characterized by a variable depth and defined as the maximum cumulative northward transport located at the transition depth between northward and southward transport, as shown in Equation (2)—was calculated and correlated with the total MHT (Kersalé et al. 2021; Meinen et al. 2013). The MOC intensity time series showed a strong correlation with the heat transport across the entire section, yielding an r value of 0.921 (Figure 8).

In this study, MHT is directly correlated with changes in MOC intensity. This corroborates both literature data and the findings of this work, indicating that variations in transported heat are linked to MOC intensity and, consequently, to the transport driven by the upper layer (Kersalé et al. 2021).

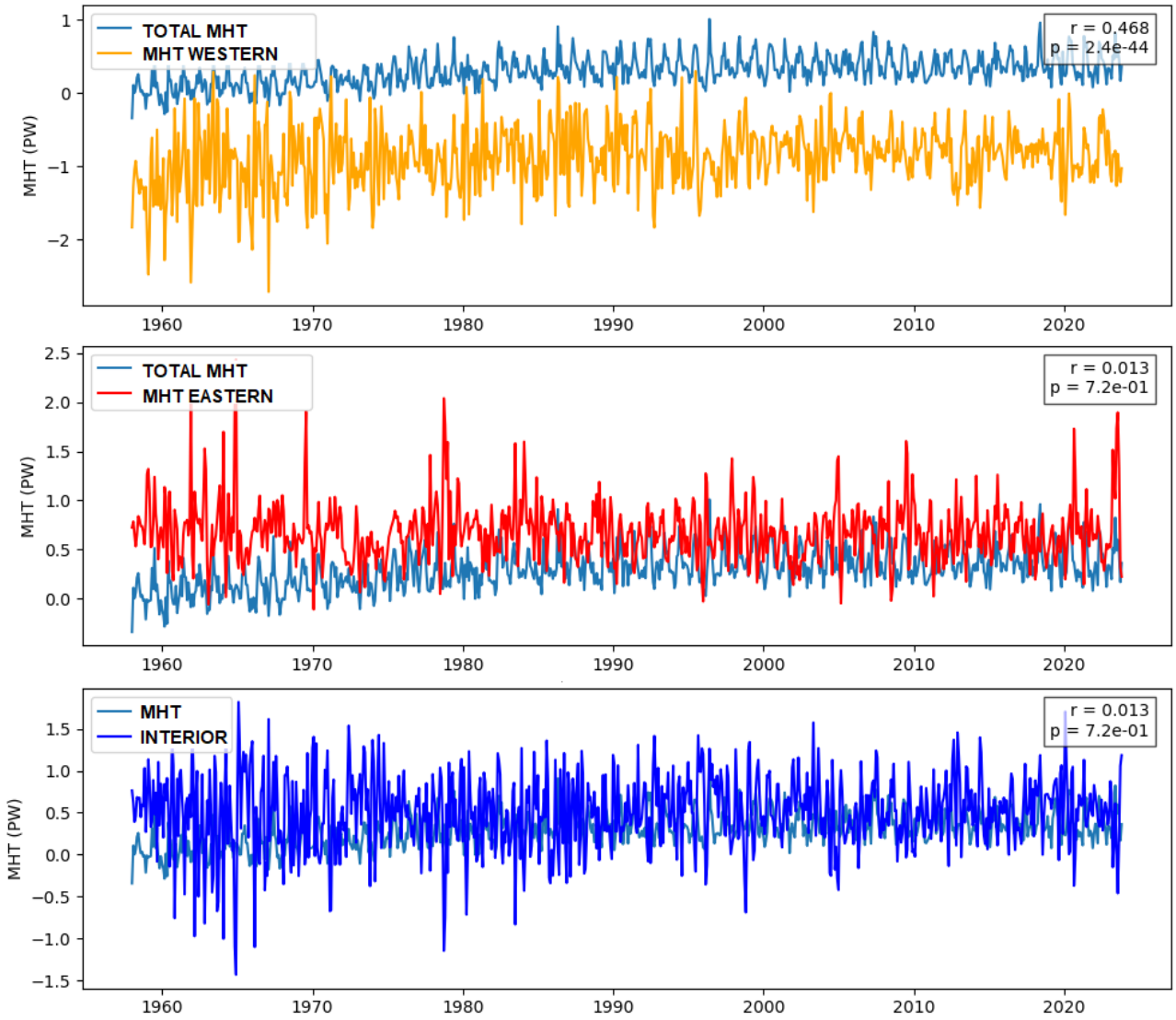


Figure 7: Comparison between total MHT and the different longitudinal regions. The linear correlation coefficient (r) is highlighted. Total calculated MHT is shown in light blue across all plots. Dark blue represents deep layer heat transport (first plot), yellow represents intermediate layer heat transport (second plot), and red represents deep-ocean layer heat transport (third plot).

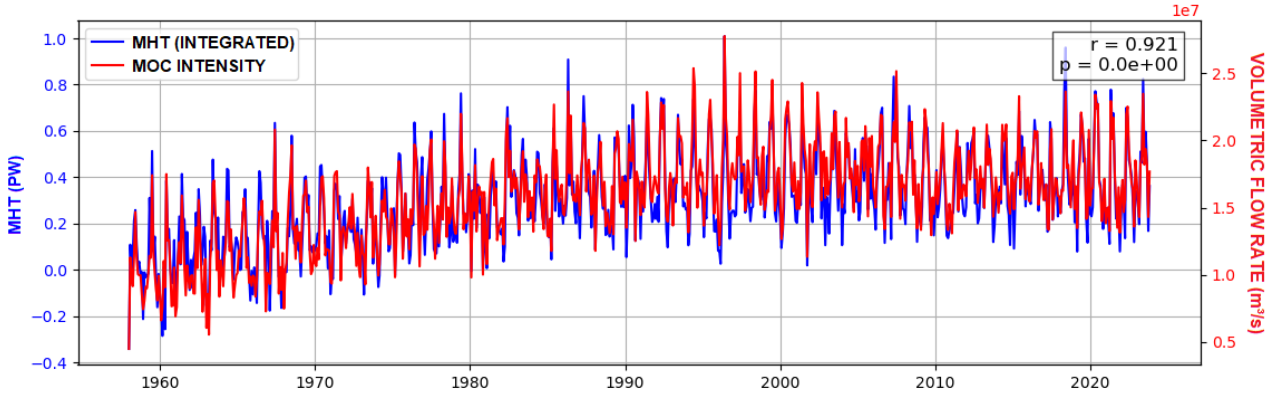


Figure 8: Correlation between MOC intensity and total basin heat transport. Total heat transport (PW) is shown in blue, and MOC intensity (m^3/s) is shown in red.

Based on the MOC intensity calculation, depth separation was achieved by determining the mean point of maximum northward AMOC transport (where the NADW originates and transport reverses southward) and the mean point of maximum southward AMOC transport (where the flow reverses northward again). The mean maximum points were identified at depths of 1159 m and 3590 m, respectively (Figure 9).

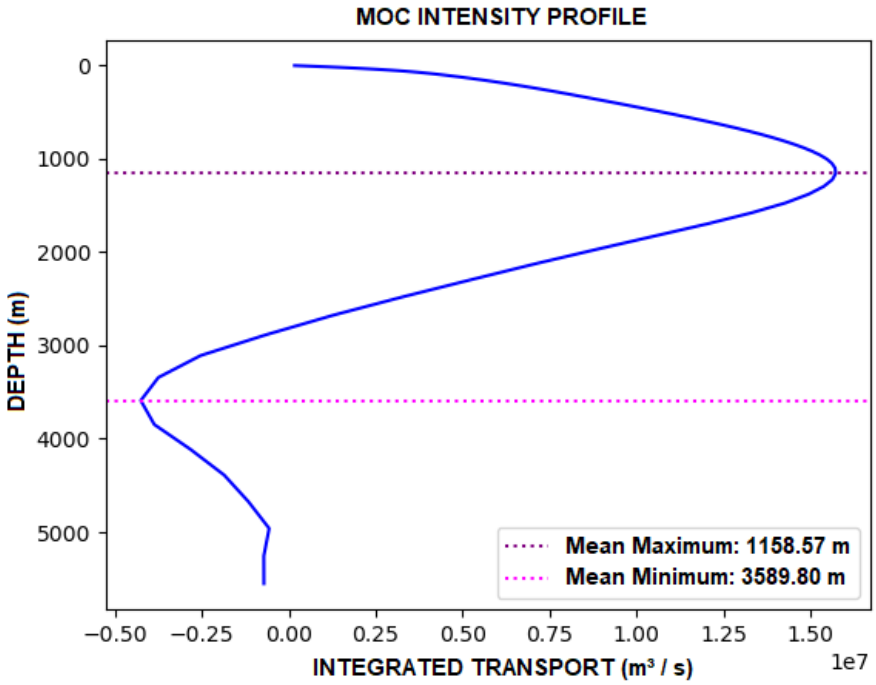


Figure 9: Mean annual AMOC profiles at 34.5°S (blue). The dashed purple line indicates the mean maximum point of northward AMOC transport, while the dashed pink line represents the mean maximum point of southward AMOC transport.

Regarding the behavior of each layer, the upper layer displays an upward trend towards the present, with an overall mean of 0.56 PW. Decadal values were 0.36 PW for the 1960s, 0.49 PW for the 1970s, 0.59 PW for the 1980s, 0.65 PW for the 1990s, 0.66 PW for the 2000s, and 0.64 PW for the 2010s. The intermediate layer exhibited a relatively constant behavior, with minimal variation around its overall mean of −0.26 PW, recording −0.25 PW for the 1960s, −0.26 PW for the 1970s, −0.27 PW for the 1980s, −0.27 PW for the 1990s, −0.27 PW for the 2000s, and −0.26 PW for the 2010s. The deep layer shows a very low heat transport magnitude in the Atlantic. Despite a slight increase towards the present starting in the 1980s, its decadal means were 1.9×10^{-3} PW for the 1960s, 2.1×10^{-3} PW for the 1970s, 1.1×10^{-3} PW for the 1980s, 1.2×10^{-3} PW for the 1990s, 2.6×10^{-3} PW for the 2000s, and 4.4×10^{-3} PW for the 2010s (Figure 10).

After confirming that MOC intensity is highly correlated with total MHT, a correlation analysis was conducted between the transport of each layer and the total cycle. The upper layer showed a behavior nearly identical to the total MHT, with a correlation coefficient of 0.994, sharing a similar variability pattern, upward trend, and temperature peaks. The intermediate layer displayed a weak inverse correlation ($r = -0.310$). The deep layer exhibited a moderate inverse correlation ($r = -0.432$), despite its low absolute magnitude of heat transport compared to the other layers (Figure 11). This phenomenon of total MHT having an opposite correlation to deep-layer MHT is also observed

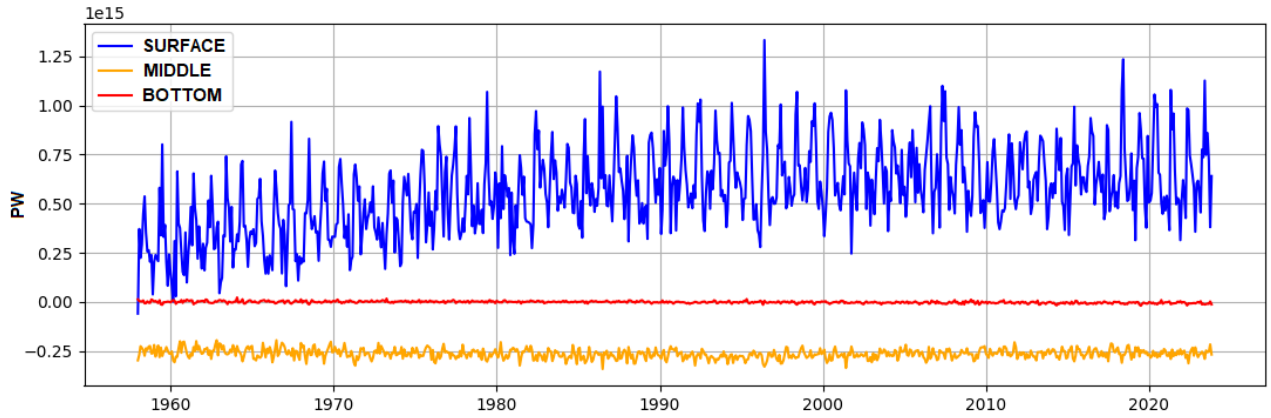


Figure 10: MHT calculation based on layer separation in the Atlantic basin: upper layer (blue), intermediate layer (yellow), and deep layer (red).

in empirical data (Kersalé et al. 2021).

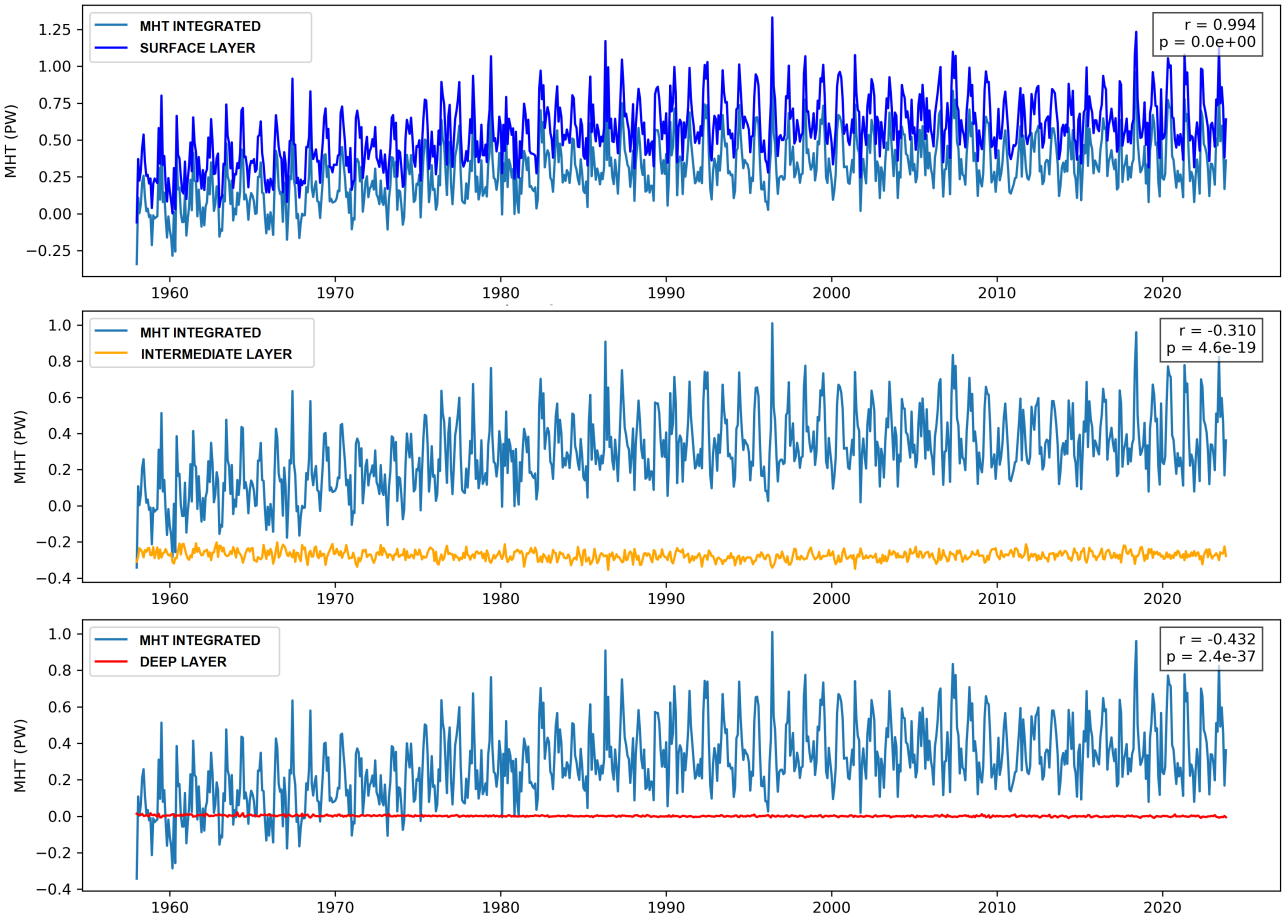


Figure 11: Comparison between total MHT and the different depth layers. The linear correlation coefficient is shown in the upper right corner of each plot. Total MHT is shown in light blue across all plots. Dark blue represents upper layer heat transport (first plot), yellow represents intermediate layer heat transport (second plot), and red represents deep layer heat transport (third plot).

Given the finding that the strongest correlation resides within the upper layer and, longitudinally, along the western margin, a targeted analysis was conducted for this specific boundary region down to a depth of 1159 m. This upper western boundary sector showed a moderate-to-high correlation with

the total transported heat, yielding an r value of 0.508 (Figure 12).

This indicates that while there is a moderate correlation with the variability of the upper western boundary (the sector encompassing the Brazil Current), the recirculation processes and dynamics of the entire upper layer as a whole are more critical than the western sector in isolation. Therefore, when discussing this circulation, the Agulhas Current system also plays an essential role in the overall heat transport of the basin.

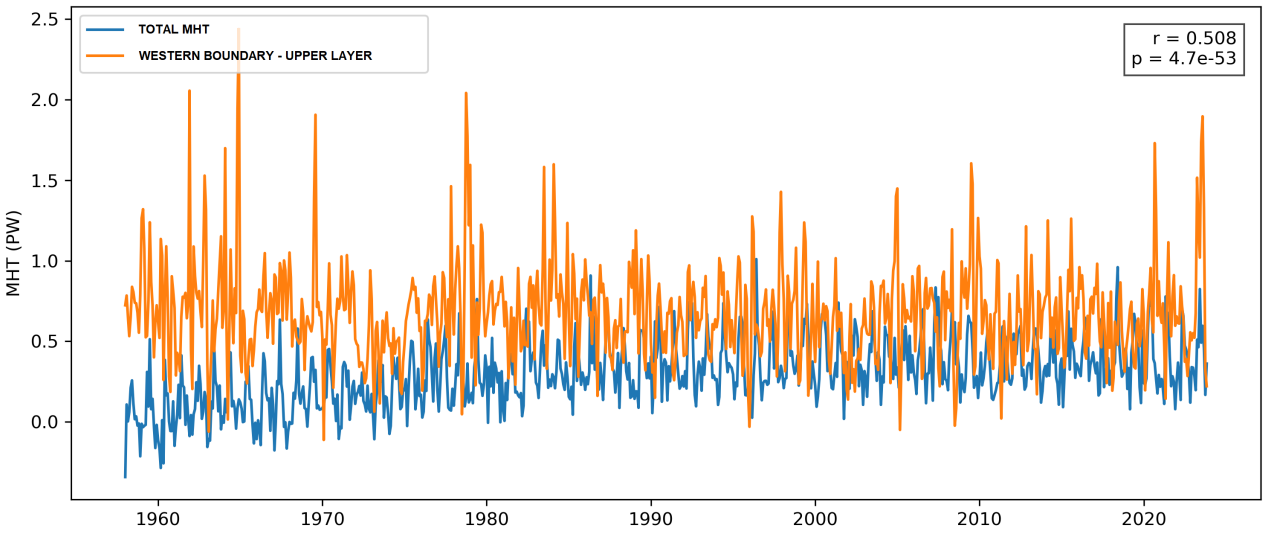


Figure 12: Correlation between the region confined between 0 and 1159 m from the continent to 49°W and the total MHT. The heat transport within this specific region is shown in orange, and the total MHT is shown in light blue.

3.5 Cycles

Following the observation that the deep layer presents an inversely proportional behavior, an analysis of the cycles within each layer was conducted. When decomposed via Fourier transform with the annual and seasonal cycles removed, the total MHT oscillation indicates two major significant cycles at 198 and 88 months, equivalent to 16.5 and 7.3 years, respectively. Other statistically relevant cycles appear, such as the 158-month cycle (13.1 years). The upper layer exhibits the same primary cycles as the total MHT, with the most significant periods at 16.5 and 7.3 years, along with the 158-month cycle (13.1 years) as a relevant signal. The intermediate layer does not show any dominant, statistically significant cycle contributing to the variability of the observed signals. Nevertheless, the same 198-month period observed in the upper layer is also prominent in deep waters (Figure 13). Therefore, these cycles in the deep and upper layers may be directly linked.

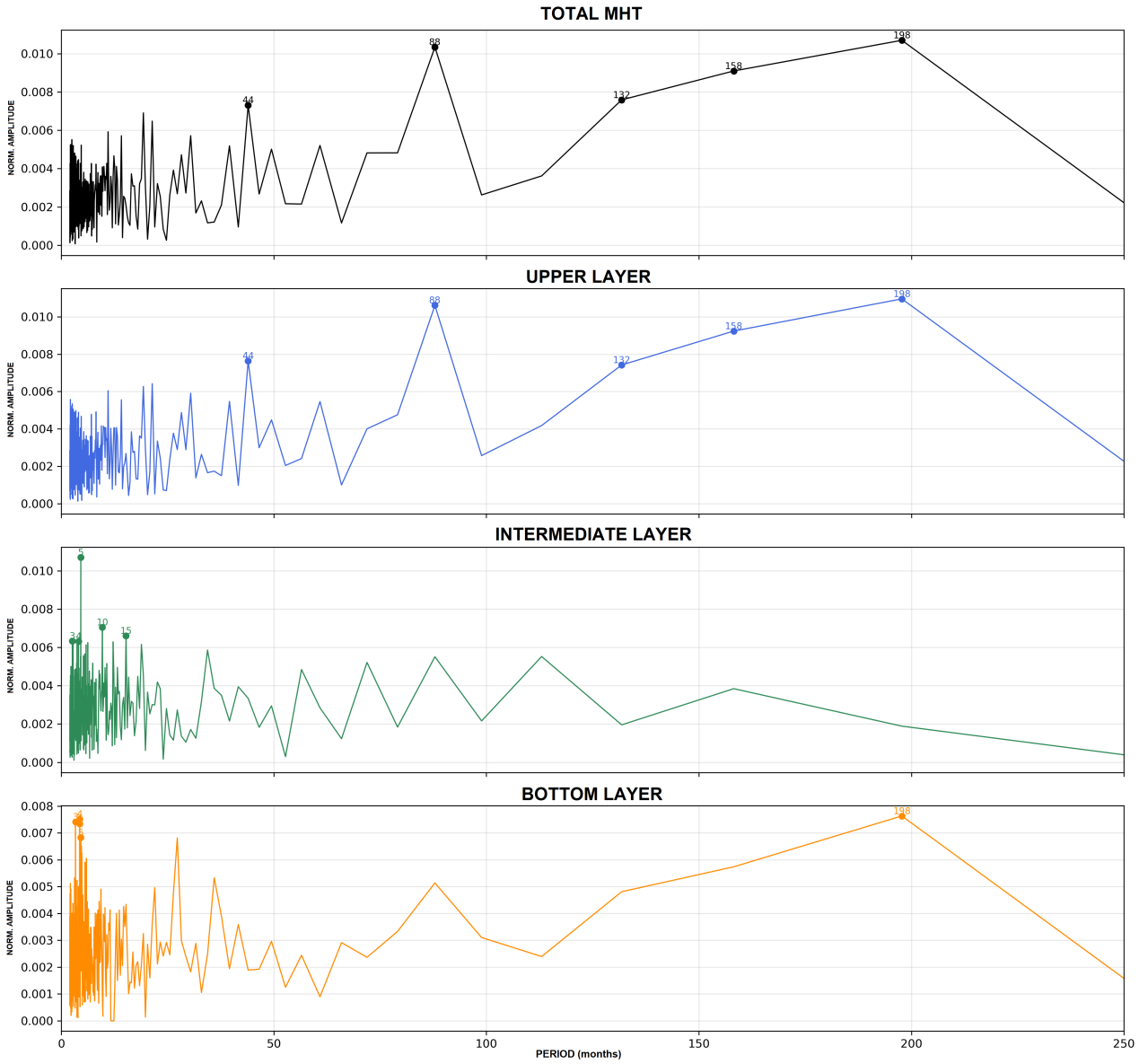


Figure 13: Fourier transform applied to the different meridional heat transports per layer: total MHT across the section (black), upper layer heat transport (blue), intermediate layer heat transport (green), and deep layer heat transport (yellow).

4 Conclusions

Over recent decades, a clear upward trend in heat transport across the Atlantic basin is observable, stabilizing near the 1990s. This increasing trend preserves seasonal interannual variability, characterized by higher heat transport during the Southern Hemisphere winter and lower values during the Southern Hemisphere summer. These fluctuations peak positively in June and reach their minimum in February, corroborating existing literature.

This seasonal variation may be linked to flux variations among boundary currents. When isolated, the upper layer shows an almost exact correlation with the total basin MHT. Furthermore, the strong correlation between MOC intensity and total MHT suggests that the velocity of these currents within the upper layers is the primary driver controlling the heat transported across the section. The

matching periodic cycles identified in both the total MHT and the upper layer transport further support this mechanism of control.

For a complete understanding of this transect, future steps should involve finer compartmentalization of the basin sectors, a detailed assessment of specific water masses, and expanded temporal coverage. Given that current data availability remains temporally constrained, long-term processes are still poorly understood. Cross-sectional comparisons within the basin factoring in upwelling and downwelling processes could further elucidate the mechanisms controlling Atlantic Ocean circulation.

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